

Sultant Of Oman
Ministry of Health
Royal Hospital



سلطنة عمان
وزارة الصحة
المستشفى السلطاني

Document Management System



Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Blood Transfusion Services

Code: BTS-POL-2-Vers.1

Effective Date: 24/09/2023

Target Review Date: 31/08/2025

Title: Policies related to transfusion of blood components

PURPOSE:

This document provides information on the policies of compatibility testing (blood group, antibody screen and cross match tests), issue of blood components to the clinical side and transfusion at bed side.

INTRODUCTION:

Safety of transfusion of blood components must be assured throughout the process of the transfusion and this process include three stages; pre-transfusion, transfusion and post-transfusion. There are key points that must be considered and strictly followed to reduce morbidity and mortality of the transfusion process. Failure to do so must trigger review and correction actions.

The most critical step is correct patient identification at both collecting the blood sample and initiation of the transfusion. Errors in the identification in either could lead to ABO incompatible transfusion haemolytic transfusion reaction which is completely preventable. It is well documented that two-thirds of the errors occurs at clinical side (wrong blood in tube incidents or transfusing the wrong patient) and one third due to testing errors.

CORRECT PATIENT IDENTIFICATION:

The following must be adhered to:

- Patient identification band must be worn by all patients receiving a blood transfusion.
- The minimum patient identifiers are - last name, first name, date of birth and patient's hospital number.
- In patients who are unable to identify themselves, for example small children, unconscious or confused patients or where there is a language barrier, verification of the patient's identity should be obtained from a parent or carer (if present at the patient's bedside) and checked against the patient identification band.

information on the patient's identification band.

- The process of transfusion must be documented in ALSHIFA system (transfusion monitoring chart)

PRE-TRANSFUSION STAGE:

Sample requirements for cross match testing:

- 4 ml EDTA (purple top) and for paediatric patient (age between 2-5 years) sample should not be less than 2ml.
- If extensive investigations (i.e. antibody investigation) blood bank will request the ward for more blood samples.
- Samples older than 7 days are not suitable for testing.
- All blood sample collection and labelling must be at the bedside as one continuous, uninterrupted event.
- In-case of failure of Al SHIFA system: Blood bank recommend hand written request form blood samples which must include a correct patient's hospital number, correctly spelled name and surname, date of birth, and time of blood sample collection.
- Samples reaching the laboratory must be will labelled; no unlabelled sample will be accepted for any investigation.
- Unlabelled samples are NOT returned to the ward or allowed to be relabelled
- All routine blood samples are to be transported to the blood bank by medical orderlies/ pneumatic system.
- For emergency blood samples, blood bank should be informed and blood sample should be given by hand to the blood bank staff and informed about the situation.
- EDTA blood samples must be in a dark blue coded plastic bag.
- Blood samples sent with spills blood in the plastic bags will not be accepted in the blood bank.
- All blood samples will be stored in blood bank refrigerator after processing for one week and will be discarded after that.

Blood group testing:

- Patient's blood group will be assigned only after two blood group determination tests (current sample compared to historical record or two samples collected on two different occasions).
- The blood group will be verified against any previous results for the patient. When verification checks against historical results reveal a discrepancy, a new sample will be obtained and tested immediately.
- If transfusion is urgently required prior to resolution of any discrepancy or confirming the blood group on a second sample in patient with absent historical record, group O packed red cells and AB plasma will be issued.
- If a discrepancy in the blood grouping is detected, the patient will be labeled as U (Unknown) blood group till discrepancy is resolved.
- For patient with no blood group history in Royal hospital Al SHIFA system, two samples are required for blood transfusion, first sample should be sent for blood group and antibody screening and the second sample should be sent when cross match is required.

Antibody screen testing:

- When irregular antibodies are detected, Antibody identification will be performed.
- Extra blood sample will be asked if the previous sample is insufficient for further testing, the ward nurse will be informed by phone to submit a new specimen. The following remark will be added to the test: insufficient specimen for antibody identification panel, please send two 4 ml EDTA samples; ward has been informed.
- If the specimen is sufficient an antibody identification panel will be automatically requested by AL SHIFA, under the same specimen number.

- Cross match will not be performed until the specificity of the antibody is determined. If the blood is required urgently then the case must be discussed with the on-call haematology (second on-call).
- Once the specificity of the alloantibodies is identified, ABO / Rh compatible units lacking the corresponding antigen specificities must be selected for the transfusion.
- **Under no circumstances should packed red cell unit be released before antibody identification is completed and antigen negative units have been fully cross matched and only a consultant haematologist can authorize any deviation from this policy.**

Cross Match:

- In no case a cross match is performed on a first-time blood group patient, a second sample must be submitted to confirm the blood group before proceeding with the cross match.
- In life threatening situation, physician can request uncross matched PRBC unit if clinically lifesaving.
- Sampling interval for cross match:

Patient transfused within	Sample to be taken not more than
3–14 days	24 h before transfusion. Please note where patients are being repeatedly transfused, a daily sample is not a requirement. At least every 72 h a new sample need to be submitted for cross match.
15–28 days	72 h before transfusion
One month to 3 months	1 week before transfusion

TRANSFUSION STAGE:

- For blood ordering, referral to documents (blood ordering for emergency cases, thalassaemia and Scheduled Operations)
- Time required for preparation of each blood component:
 - Red blood cells cross-match will require 2 hours if patients' blood does not contain any abnormal antibodies.
 - FFP and Cryoprecipitate will require 30 minutes for three units to be prepared
 - Platelet request will need at least 15 minutes to be ready.
- Blood bank staff will allocate the availability of the blood component in Al SHIFA system.
- Ward staff should check blood availability in the computer before sending well trained staff for blood product collection. No phone calls will be accepted by blood bank for enquiring of routine blood availability.
- No blood component will be released to the wards without the following:
 - Patient hospital number sticker
 - The required blood component is clearly mentioned on the patient sticker.
 - A well labeled clean cool box container

Bedside checks pre-collection of blood components

- The reason for the transfusion has been explained to patient (and/or for paediatric patients those with parental responsibility) and consent obtained and documented in the patient's clinical records.
- There is appropriate and patent intravenous access.
- There should be trained and competent staff available for the duration of the transfusion.
- The patient's baseline (vitals) observations have been completed.

Transfusion at bed side:

- Blood components received from blood bank must be checked by a competent nursing staff and all the important documents should be signed.
- Blood components must be transfused within 30 minutes of receiving it from blood bank, or return to blood bank immediately.
- Blood component must not be stored in normal or drug fridges in any case, if proper blood fridge is not available in the ward, blood must be returned to blood bank.
- Transfusion must only take place when there are enough staff available to monitor the patient and when the patient can be readily observed.
- Overnight transfusions should be avoided unless clinically essential.
- Strictly follow Royal hospital nursing department policy on administration of blood components policy.
- It is preferred at all-times to assure the presence of a physician in the ward/ on call duty room during the transfusion process.
- No patient should be left unattended while transfusion is on.
- Any adverse reaction must be reported and transfusion stopped immediately.
- Transfusion process of one unit of PRBC should not take **more than 4 hours**.

POST TRANSFUSION STAGE:

- The time each blood unit was completed should be recorded.
- If a further blood component unit is prescribed, repeat the administration and patient identity check for each unit.
- If no further units are prescribed, remove the blood administration set.
- If any other IV fluids are prescribed, they should be administered using a new IV administration set appropriate for that solution.
- It is not necessary to 'flush' the blood administration set after transfusion. However, if this is done, only isotonic (0.9%) saline should be used.

Written By: Dr Halima Al Balushi

Checked By: Dr Muna Al Riyami

Authorized by: Sabria Nasser Khalfan Al Hashami

Review Details

	Review BY	Review Date
1	Dr Sabriya Al Hashami	24/01/2024

[Back](#)


© Copyright 2018 reserved 'Royal Hospital-
Information Technology Department'